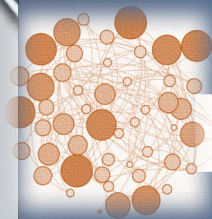
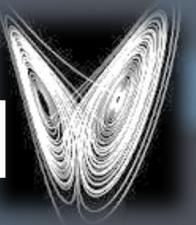


Lecture 1



Lab For Networks And Nonlinear Dynamics
(LN²D)



Fast and Small - Subhadip Mitra

The Special Theory of Relativity and the Elementary World

Slow and Big - Diganta Das

Cosmology

Patterns in the Middle - Chittaranjan Hens

Statistical Mechanics

Title of the Course : The Universe Across Scales

*The aim of the course is to present a broad overview of physics across different scales—from the quantum mechanical world of the elementary particles to gravity-controlled large-scale structures through physics at the everyday scales. To teach the students **simple calculations and estimations to appreciate the beautiful rules** that nature seems to follow at different scales and the **emerging patterns**.*

Unit-1: Fast and small

- 1. Special Theory of Relativity: time dilation, length contraction, Lorentz transformation, and the mass-energy relation [2]*
- 2. A brief introduction to the core ideas of Quantum Mechanics: the double-slit experiment, the Schroedinger equation, wave-function, the basic postulates, and the particle-in-a-box problem [3]*
- 3. Elementary particles, fundamental interactions, composite states: nucleons, atoms, molecules [3]*

Title of the Course : The Universe Across Scales

Unit-3: Slow and big

- 7. Large-scale structure of the Universe, a brief history [1]*
- 8. Olber's paradox, Isotropy and homogeneity, Hubble's law [1]*
- 9. The Universe: according to Newton and Einstein [2]*
- 10. Dynamics of the expanding Universe: Friedmann equations [2]*
- 11. Cosmic Microwave Background, Big Bang Nucleosynthesis and inflation [2]*

Title of the Course : The Universe Across Scales

- *4. Statistical Physics in Brief*
- *5. Universal scaling in networks, Fractals in nature—from the sub-cellular level to social interactions (analysis with real networks)*
- *6. Collective motion of self-propelled particles/swarmmalators: Flocking of birds and bacterial clusters. Scale-invariance property under different scales: from microorganisms to large ecological systems*

Dynamical Processes on Complex Networks by Alain Barrat, Marc Barthelemy, Alessandro Vespignani
Introduction to Computational Physics, Lecture of Prof. H. J. Herrmann

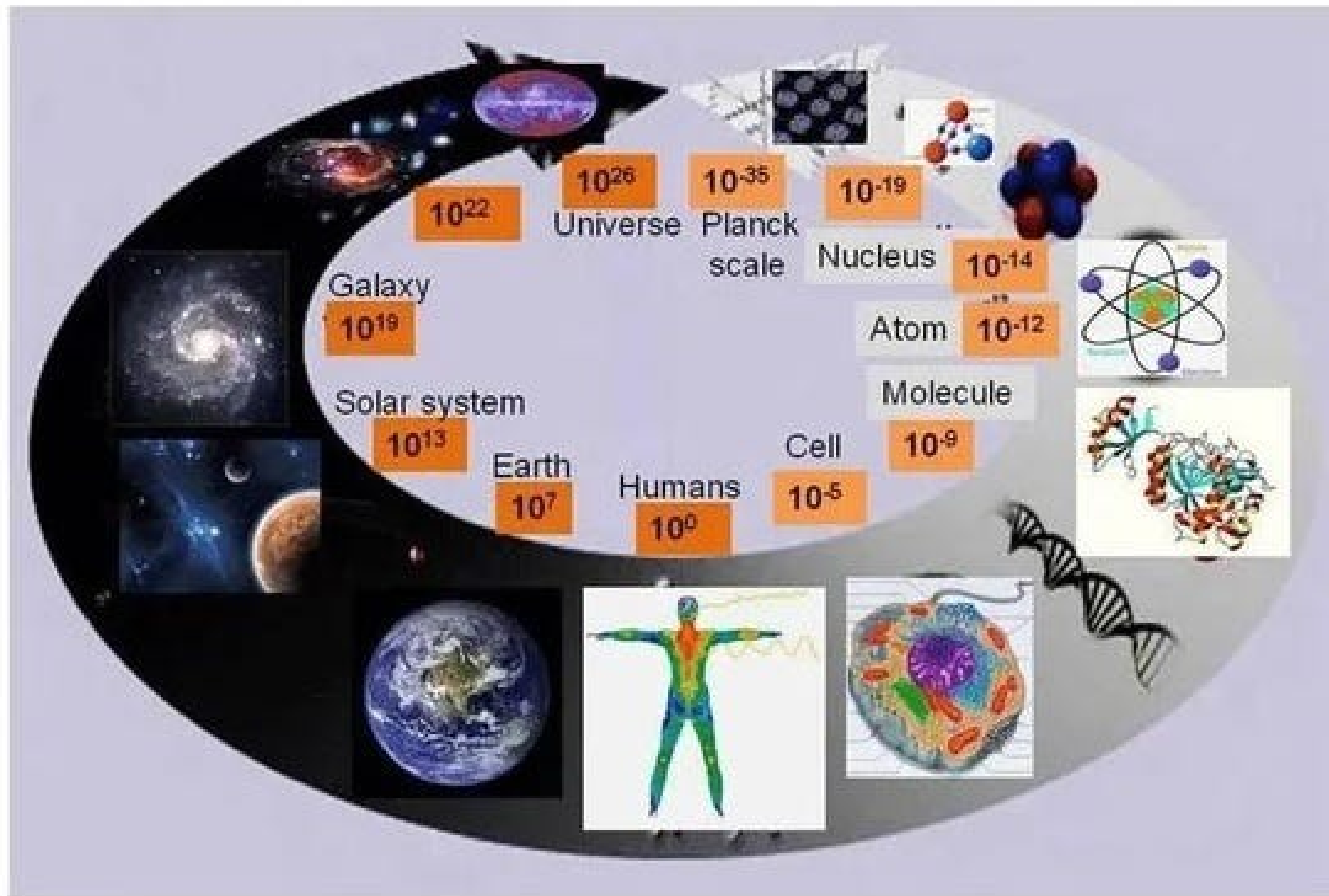
Percolation and Ising model

Scaling the Universe from the macro to micro level in meters

Fast

Slow

Patte

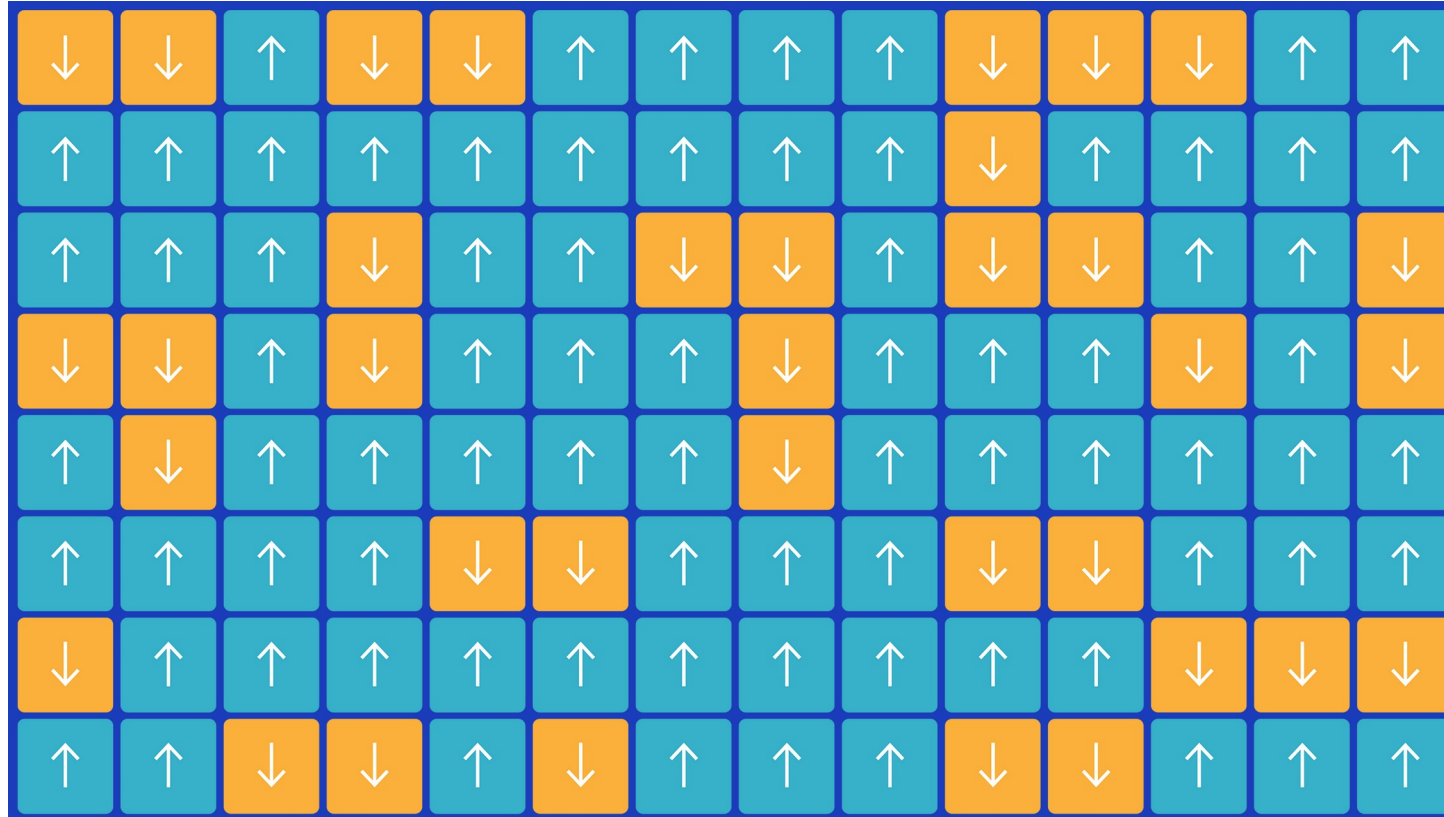


tary World

Cosmology

Mechanics

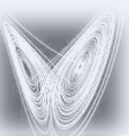
Magnetization



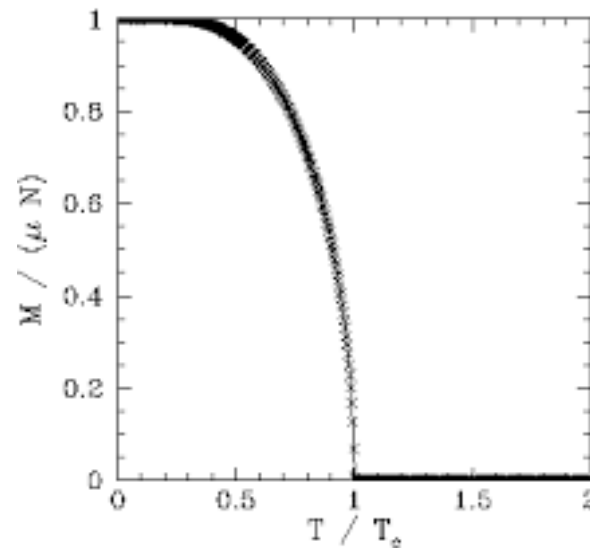
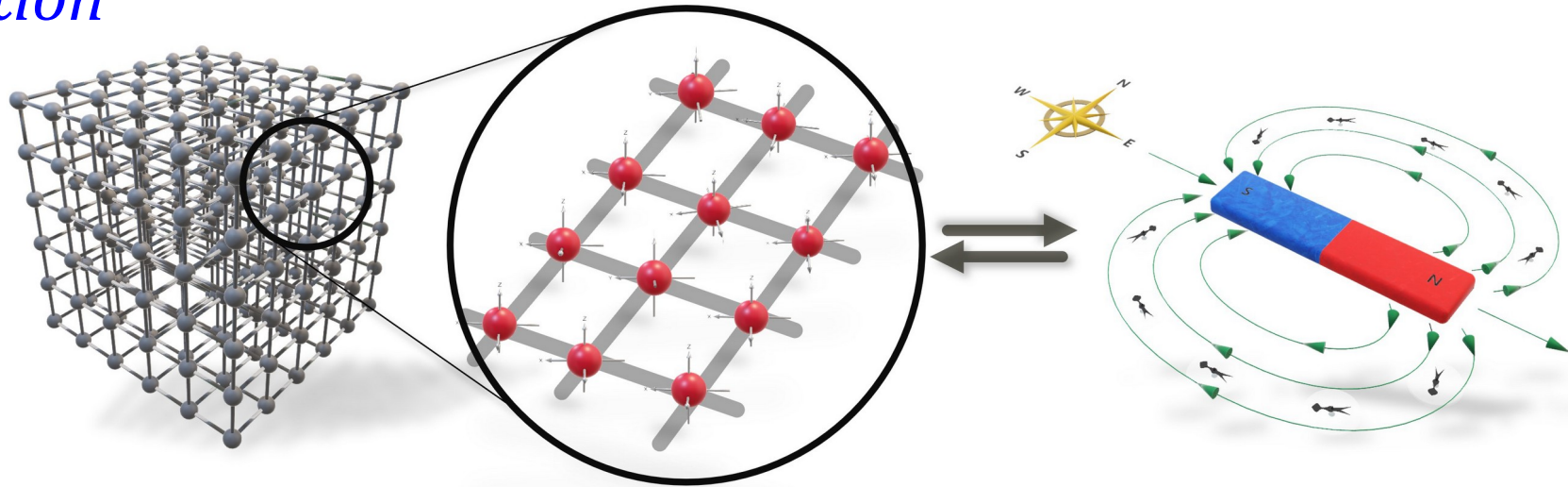
<https://www.britannica.com/video/dependence-properties-temperature/-203977>



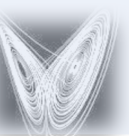
Lab For Networks And Nonlinear Dynamics
(LN²D)



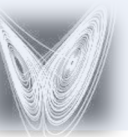
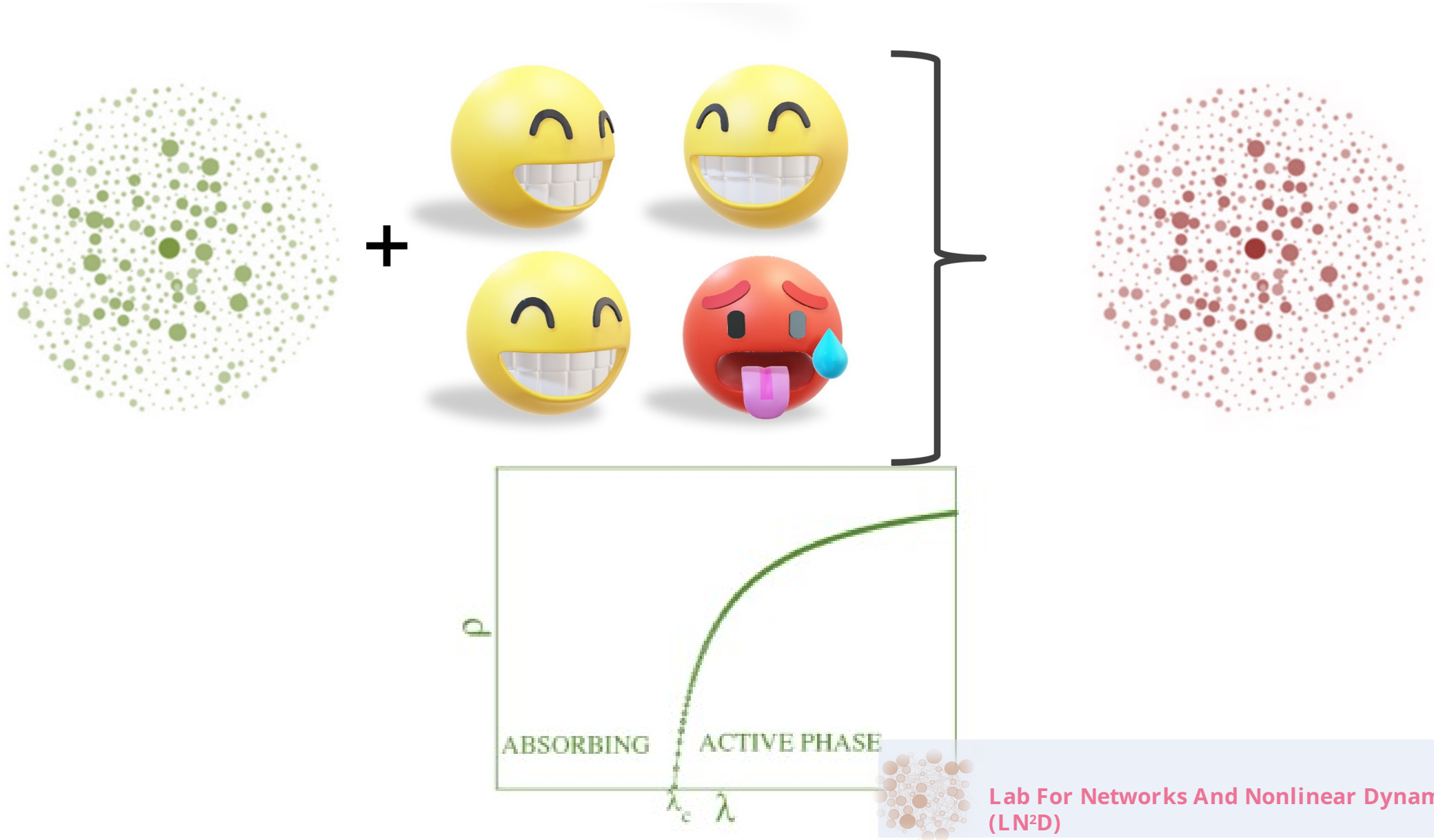
Magnetization



μ is the atomic magnetic moment. M is the net magnetization



Infection/ rumors



Percolation

A representative question (and the source of the name) is as follows. Assume that some liquid is poured on top of some porous material.

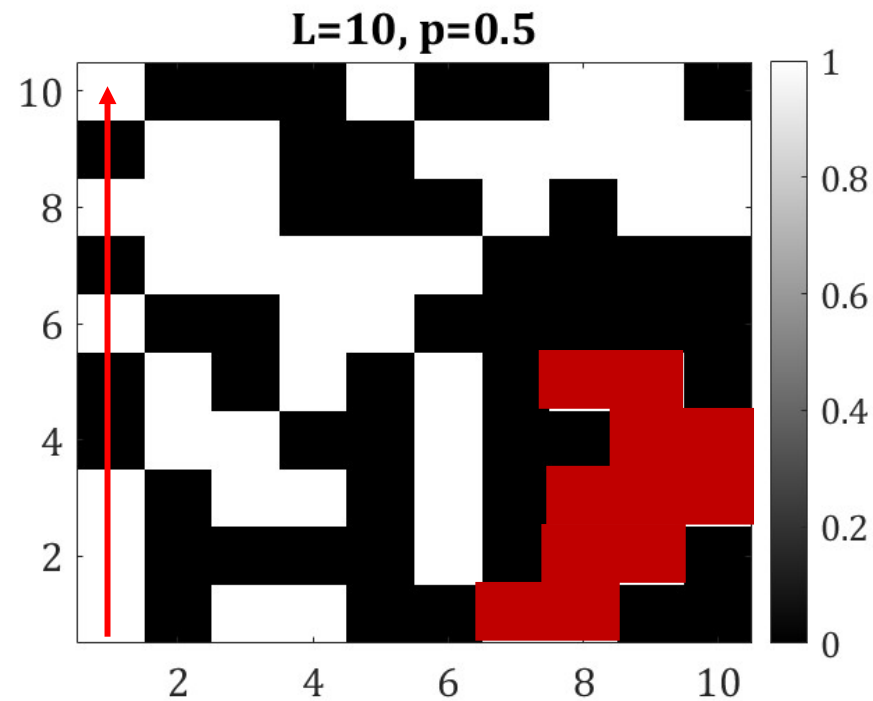
Will the liquid be able to make its way from hole to hole and reach the bottom?

This physical question is modelled mathematically as a three-dimensional network of $n \times n \times n$ vertices, usually called "sites", in which the edge or "bonds" between each two neighbors may be **open (allowing the liquid through)** with probability p , or closed with probability $1 - p$, and they are assumed to be independent.

Therefore, for a given p , what is the probability that an open path (meaning a path, each of whose links is an "open" bond) exists from the top to the bottom?

Percolation in 2-d

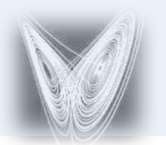
```
L = 20;  
p = 0.5;  
z = rand(L,L);  
m = z < p;  
imagesc(m);
```

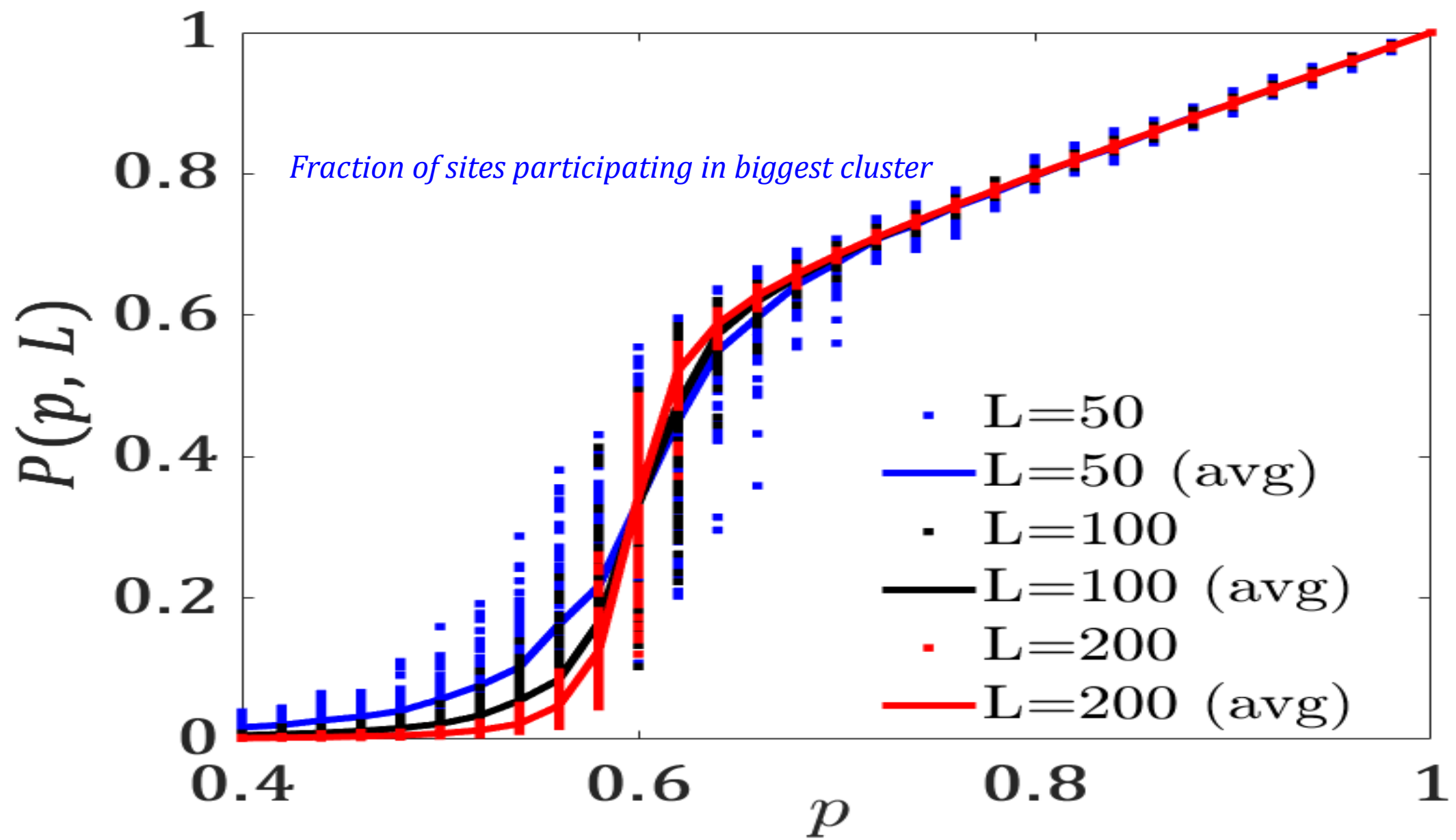


Marking the clusters

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

1	1	0	6	6	0	0	10	10	0	0
2	1	0	0	0	0	8	0	10	10	0
3	1	0	5	5	0	8	0	10	10	10
4	0	5	5	0	0	8	0	0	10	10
5	0	5	0	3	0	8	0	10	10	0
6	2	0	0	3	3	0	0	0	0	0
7	0	3	3	3	3	3	0	0	0	0
8	3	3	3	0	0	0	9	0	9	9
9	0	3	3	0	0	9	9	9	9	9
10	4	0	0	0	7	0	0	9	9	0



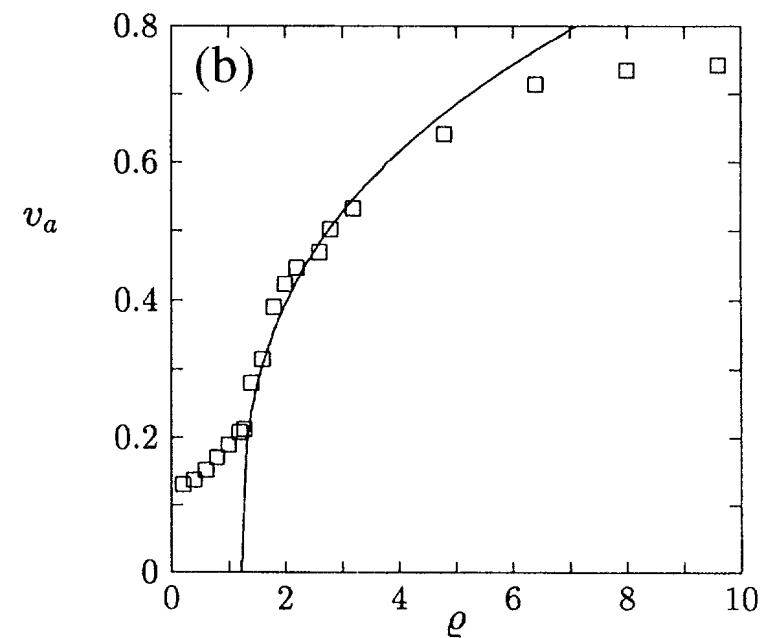
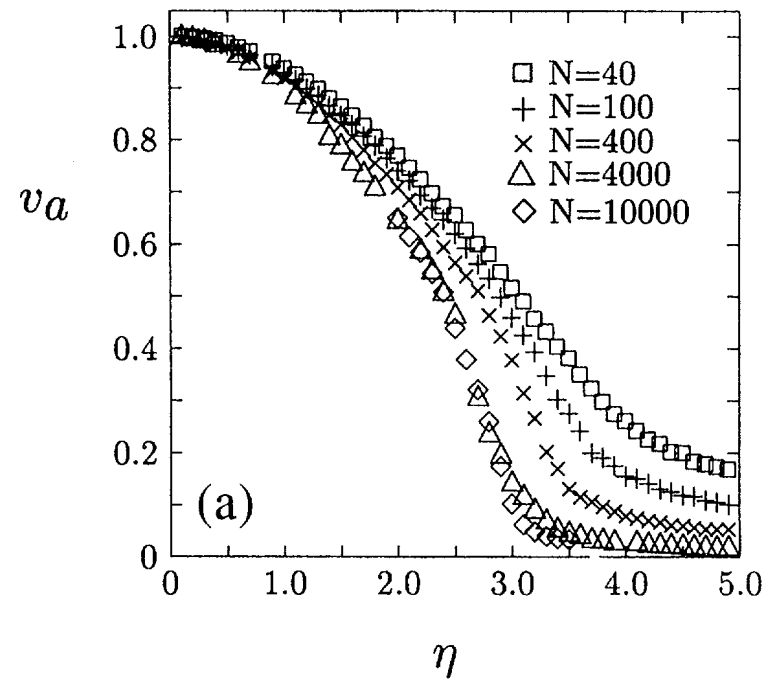
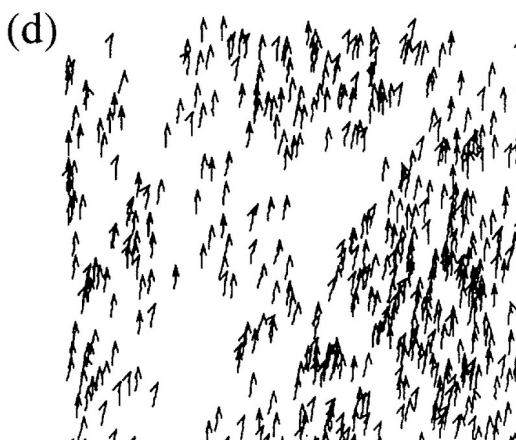
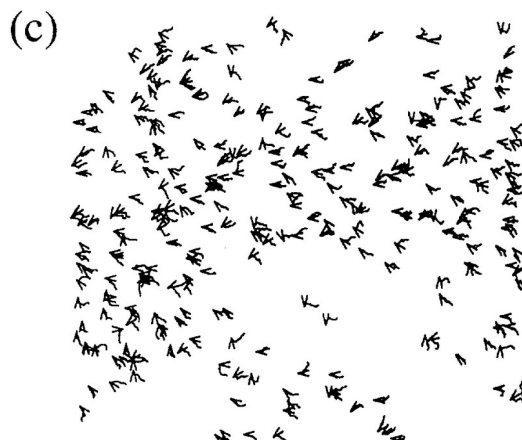
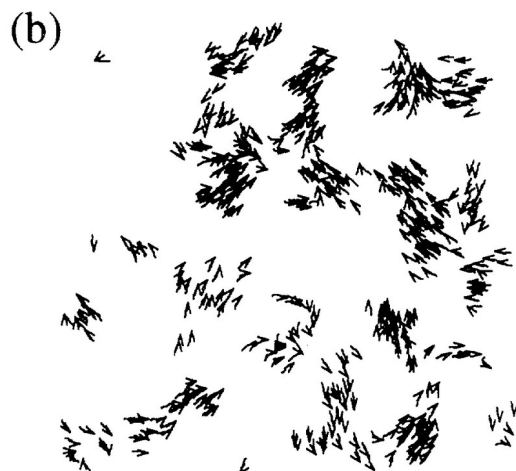
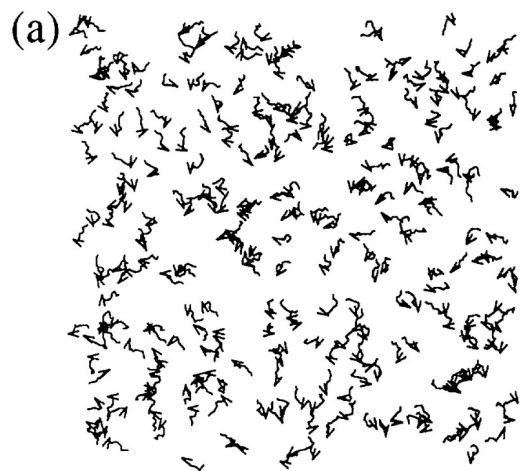


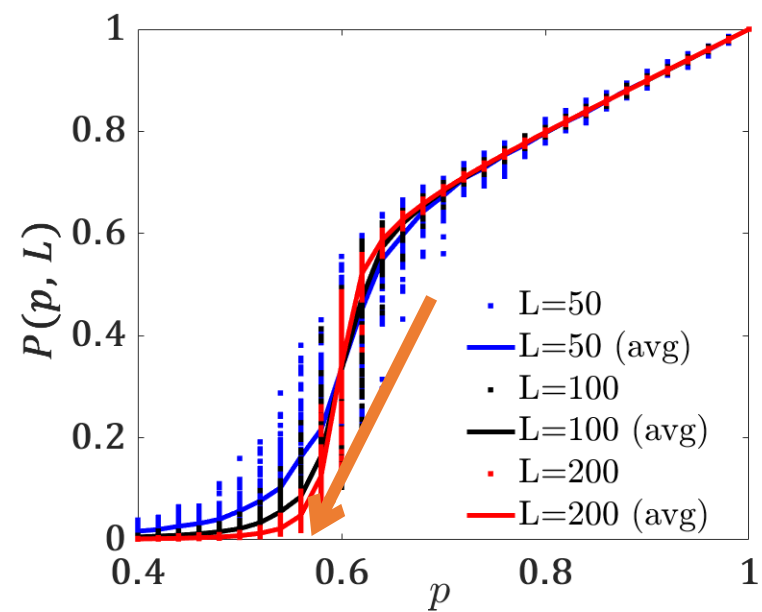
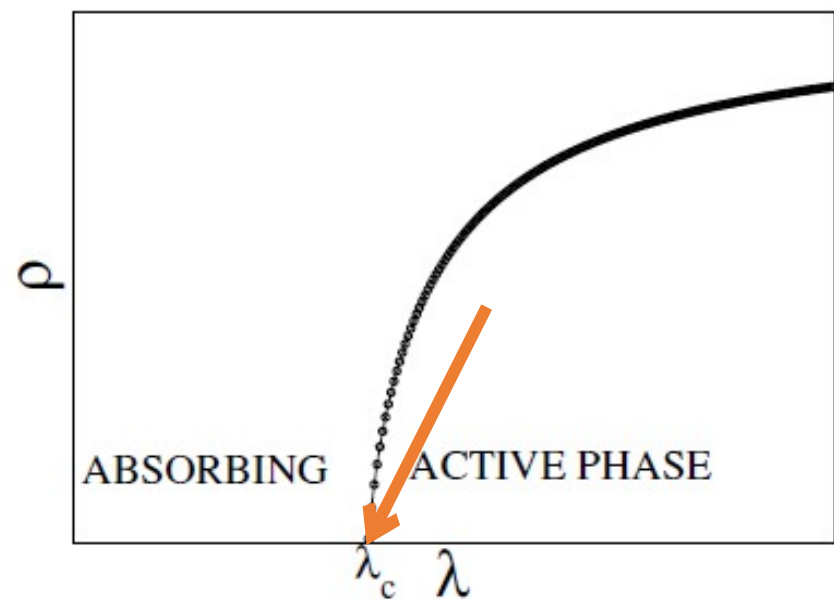
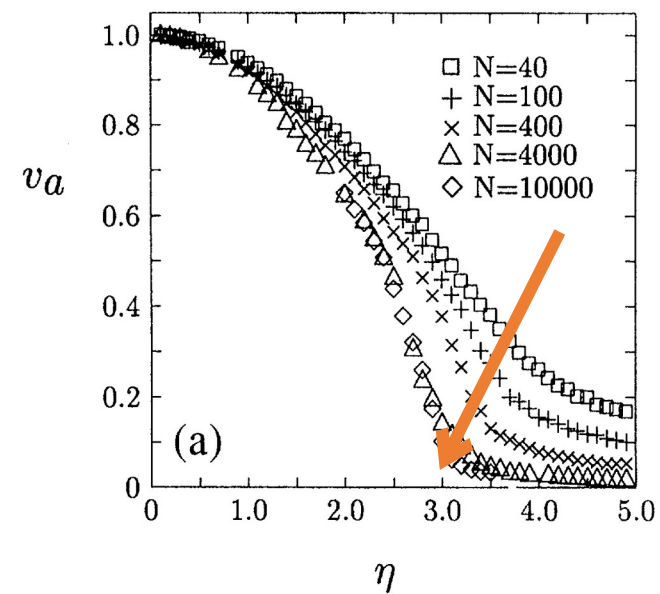
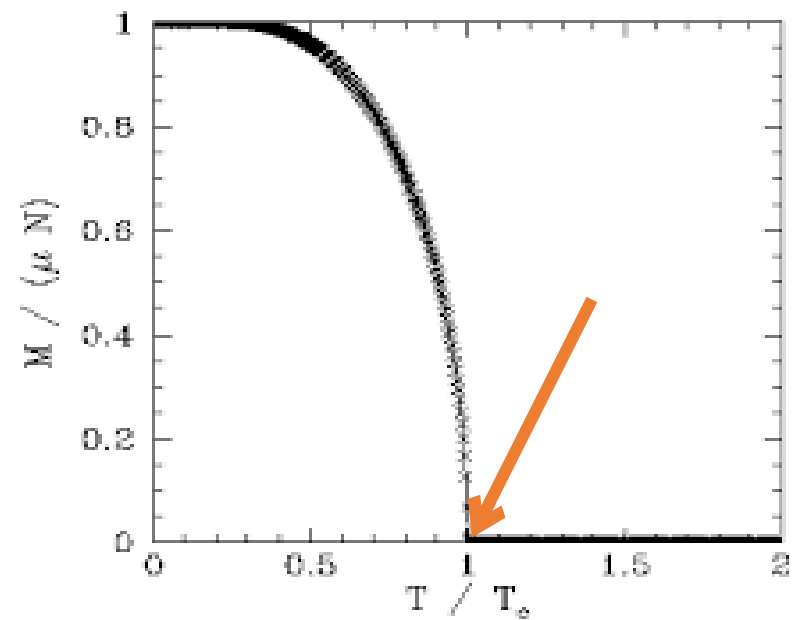
Why do birds flock together?



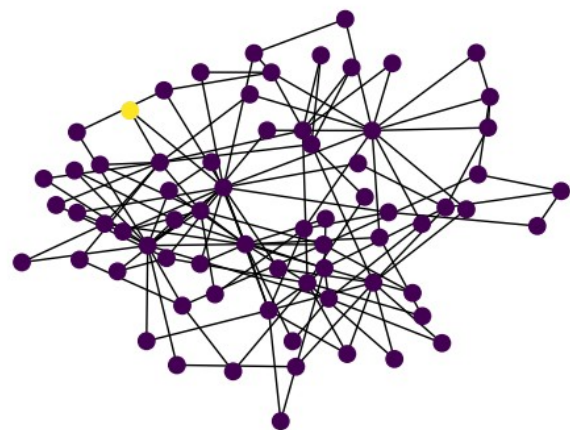
How do flocks of birds fly so close together without colliding?

Why do birds flock together?

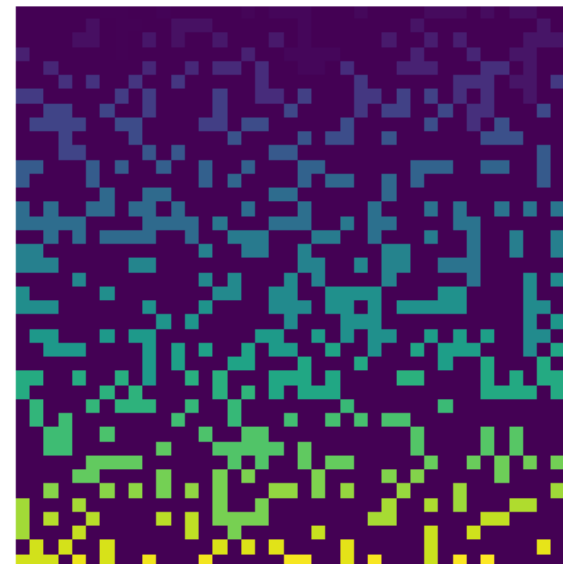




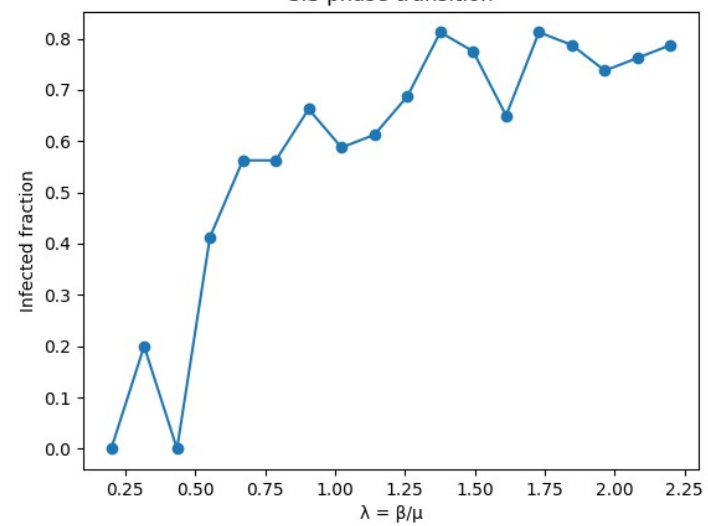
SIS network state $\lambda = 0.20$



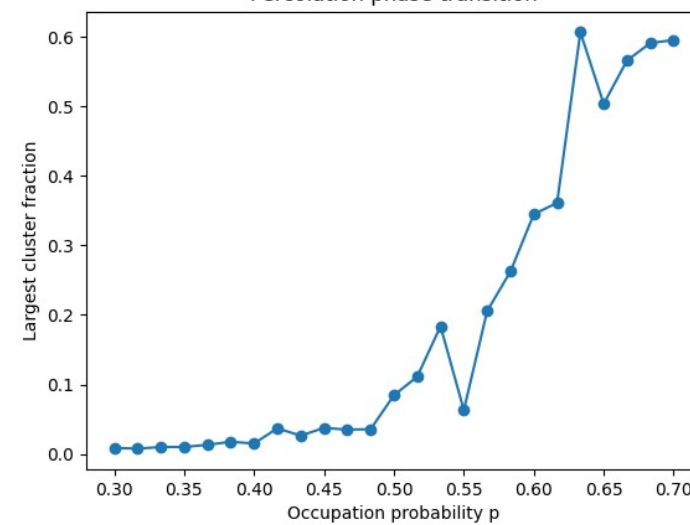
Percolation clusters $p=0.30$



SIS phase transition



Percolation phase transition



Model	Control parameter			Order parameter
SIS epidemic	Model	Control parameter	Order parameter	<i>infected fraction</i>
	SIS epidemic	infection rate β/μ	infected fraction	
	Percolation	occupation probability p	giant cluster size	
	Ising	temperature T	magnetization	
Percolation	Model	Control parameter	Order parameter	<i>giant cluster size</i>
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Scaling



Scaling BTW model



*The model is a **cellular automaton**. In its original formulation, each site on a finite grid has an associated value that corresponds to the slope of the pile. This slope builds up as "grains of sand" (or "chips") are randomly placed onto the pile, until the slope exceeds a specific threshold value at which time that site collapses transferring sand into the adjacent sites, increasing their slope.*

Bak, Tang, and Wiesenfeld considered process of successive random placement of sand grains on the grid; each such placement of sand at a particular site may have no effect, or it may cause a cascading reaction that will affect many sites.

Scaling

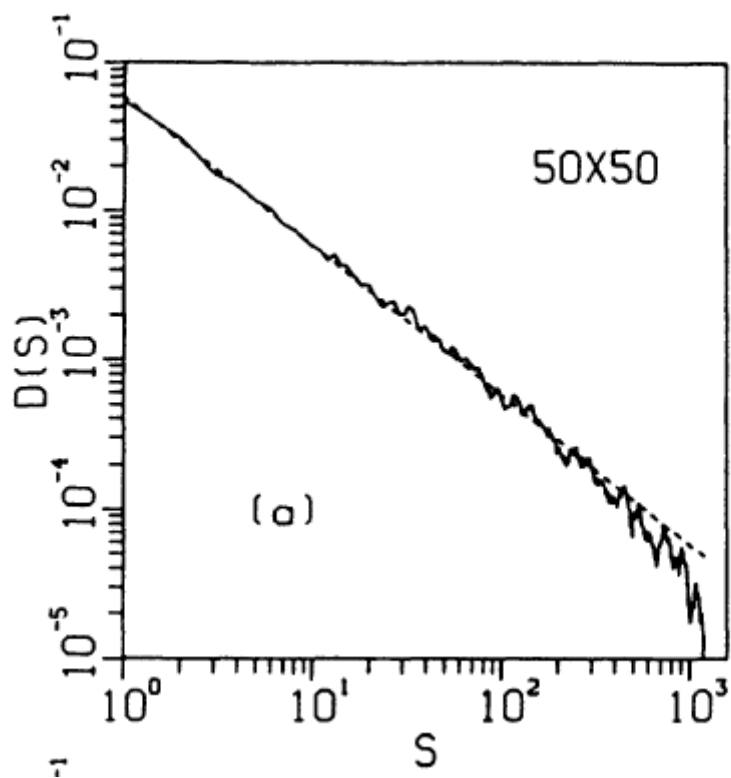
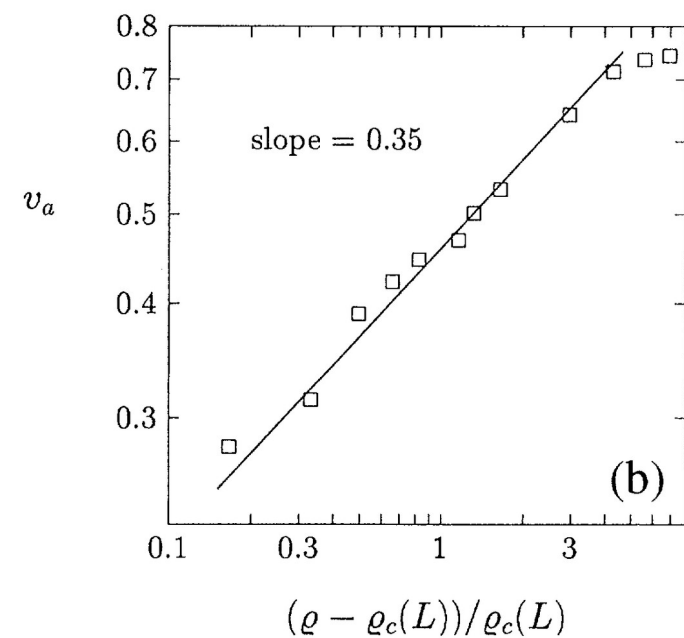
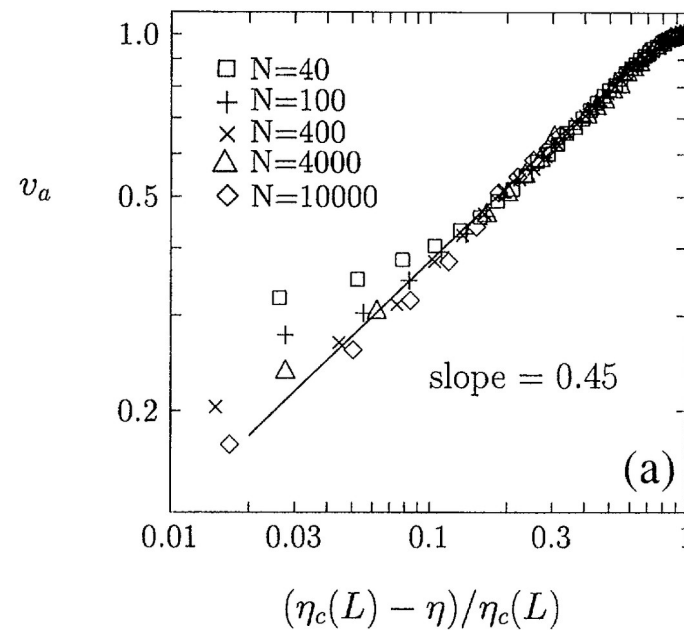
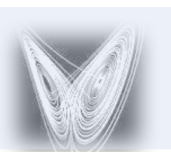


FIG. 3. Distribution of cluster sizes at criticality in two and three dimensions computed as described in the text. The data have been coarse grained. (a) 50×50 array, averaged over 200 samples. The dashed line is a straight line with slope -1.0 ; (b)



SIS Model (Homogeneous mixing, average neighbors $\langle k \rangle$)

- Most pathogens are eventually defeated by the immune system or by treatment. To capture this fact, we need to allow the infected individuals to recover, ceasing to spread the disease.
- With that we arrive at the so-called *SIS model*, which has the same two states as the SI model, susceptible and infected. The difference is that now infected individuals recover at a fixed rate μ , becoming susceptible again.
- The equation describing the dynamics of this model is an extension of SI equation



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$$\frac{di}{dt} = \beta \langle k \rangle i(1 - i) - \mu i$$

$$i = \left(1 - \frac{\mu}{\beta \langle k \rangle}\right) \frac{C e^{(\beta \langle k \rangle - \mu)t}}{1 + C e^{(\beta \langle k \rangle - \mu)t}}$$

$$i_0 = i(t = 0) \text{ gives } C = i_0 / (1 - i_0 - \mu / \beta \langle k \rangle).$$

SIS Model (Homogeneous mixing, average neighbors $N - 1$)

$$i = \frac{\frac{\beta - \gamma}{\beta}}{1 + \left(\frac{\beta - \gamma}{\beta i_0} - 1\right) e^{-(\beta - \gamma)t}}$$

$$i_{ee} = \frac{\beta - \gamma}{\beta} \text{ endemic equilibrium, } t \rightarrow \infty$$

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Characteristic time

$$\tau = \frac{1}{\mu(R_0 - 1)}$$

where

$$R_0 = \frac{\beta \langle k \rangle}{\mu}$$

is the *basic reproductive number*. It represents the average number of susceptible individuals infected by an infected individual during its infectious period in a fully susceptible population. In other words, R_0 is the number of new infections each infected individual causes under ideal circumstances.

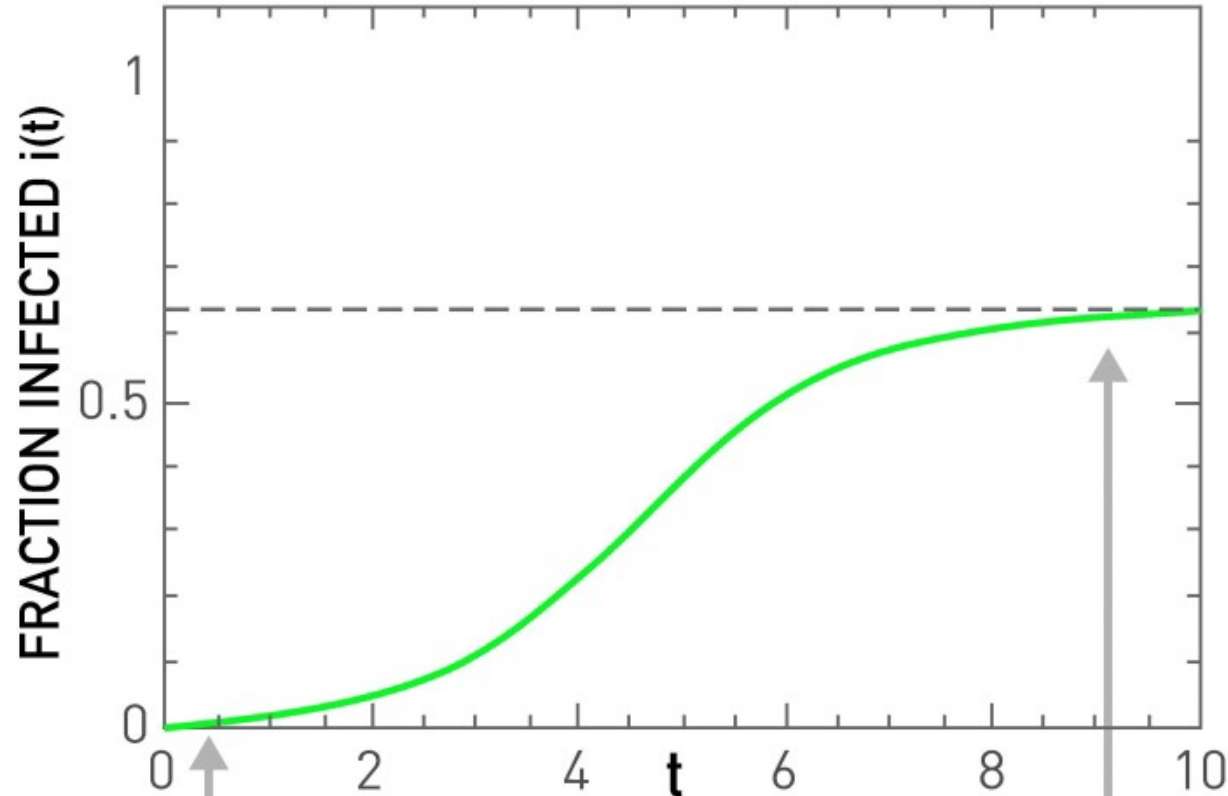
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SIS Model (Homogeneous mixing, average neighbors $\langle k \rangle$)

- Λ
- t
- r
- v
- a
- i



exponential outbreak

If i is small,
 $i \approx i_0 e^{(\beta \langle k \rangle t - \mu)t}$

endemic state

$$i(\infty) = 1 - \frac{\mu}{\beta \langle k \rangle}$$

immune system or by infected individuals to

as the same two states
 e is that now infected
 le again.

average number of susceptible
 its infectious period in a fully
 ber of new infections each

.

SIS Model (Homogeneous mixing, average neighbors $\langle k \rangle$)

Input: N , $k=10$, β , μ , T

1. Create random network G with N nodes
2. Set connection probability $p = k/(N-1)$
3. Generate edges with probability p

4. Initialize node states
 randomly infect I_0 nodes
 rest susceptible

5. For $t = 1$ to T

$\text{new_state} = \text{current_state}$

 For each node i

 if node i is Infected:
 with probability μ :
 $\text{new_state}[i] = \text{Susceptible}$

 if node i is Susceptible:
 $m = \text{number of infected neighbors}$
 $\text{infection_prob} = 1 - (1-\beta)^m$

 with probability infection_prob :
 $\text{new_state}[i] = \text{Infected}$

 update all node states

 record number of infected nodes

6. Output infection time series

- *Statistical Physics in Brief*
- *Universal scaling in networks, Fractals in nature—from the sub-cellular level to social interactions (analysis with real networks)*
- *Collective motion of self-propelled particles/swarmmalators: Flocking of birds and bacterial clusters. Scale-invariance property under different scales: from microorganisms to large ecological systems*

Dynamical Processes on Complex Networks by Alain Barrat, Marc Barthelemy, Alessandro Vespignani 4. Introduction to Computational Physics, Lecture of Prof. H. J. Herrmann

- SOC: Sand-pile model
- Power law: Data, scale-free graph, percolation in critical points, sand-pile model cluster size distribution
- Ising model: critical onset of magnetization
- Fractal and chaos

- Cellular Automata (Hiroki Samaya)
- Flocking of birds analysis

<https://www.complexity-explorables.org/explorables/horde-of-the-flies/>

https://web.mit.edu/8.334/www/grades/projects/projects10/Hernandez-Lopez-Rogelio/dynamics_2.html